

A TEAM BY VU SQUAD

**CS-301 MCQS+SHORT QUESTION
ANSWER FOR FINAL TERM**

VUSQUAD134@GMAIL.COM

03137248280

SUBSCRIBE MY VU SQUAD CHANNEL

**[https://www.youtube.com/channel/UC29f
OQkUyTKY0QzOQ5rPZAAQ](https://www.youtube.com/channel/UC29fOQkUyTKY0QzOQ5rPZAAQ)**

MOHSHIN

Question # 1 of 10 (Start time: 10:35:18 PM, 08 March 2022)

Binary Search can be categorized into which of the following?

Select the correct option

☐	Dynamic programming
☐	Divide and conquer
☐	Greedy algorithm
☐	Brute Force technique

CORRECT ANS
MOHSIN CHINIOTI
03137248280

MOHSIN

Question # 2 of 10 (Start time: 10:35:56 PM, 08 March 2022)

During the union by size method, all the array element initialized to -1 shows;

Select the correct option

☐	Initial condition of tree	CORRECT ANS MOHSIN CHINIOTI 03137248280
☐	Every tree has two elements each.	
☐	None of given.	
☐	Tree is complete now	

MOHSIN

Question # 3 of 10 (Start time: 10:37:48 PM, 08 March 2022)

The total number of nodes on 10th level of a perfect binary tree are :

Select the correct option

☐	Can't be determined	CORRECT ANS MOHSIN CHINIOTI 03137248280
☐	256	
☐	1024	
☐	512	

MOHSIN

Question # 4 of 10 (Start time: 10:39:38 PM, 08 March 2022)

The total number of nodes on 5th level of a perfect binary tree are :

Select the correct option

☐	32	CORRECT ANS MOHSIN CHINIOTI 03137248280
☐	31	
☐	16	
☐	15	

MOHSIN C

Question # 5 of 10 (Start time: 10:41:42 PM, 08 March 2022)

Which of the following properties are satisfied by Equivalence relationship?

Select the correct option

☐	Only Reflexive and Symmetric
☐	Reflexive , Symmetric and Transitive
☐	Only Reflexive and Transitive
☐	Only Symmetric and transitive

CORRECT ANS
MOHSIN CHINIOTI
03137248280

MOHSIN

Question # 6 of 10 (Start time: 10:42:13 PM, 08 March 2022)

Suppose a hash function returns 6 for the given value. At which index of array the value will be saved?

Select the correct option

☐	0	
☐	5	
☐	6	CORRECT ANS MOHSIN CHINIOTI 03137248280
☐	7	

MOHSIN

Question # 7 of 10 (Start time: 10:43:39 PM, 08 March 2022)

If Ahmed is cousin of Ali and Ali is cousin of Asad then Ahmed is also cousin of Asad. This statement has the following property

Select the correct option

☐	Transitivity	CORRECT ANS MOHSIN CHINIOTI 03137248280
☐	Symmetry	
☐	All of the above	
☐	Reflexivity	

MOHSIN

Question # 8 of 10 (Start time: 10:44:36 PM, 08 March 2022)

The array in binary search is sub divided _____.

Select the correct option

☐	Once
☐	Until a sublist is no more divisible
☐	N time
☐	Twice

CORRECT ANS
MOHSIN CHINIOTI
03137248280

MOHSIN

Question # 9 of 10 (Start time: 10:44:55 PM, 08 March 2022)

Suppose there are 100 elements in an equivalence class, so initially there will be 100 trees. The collection of these trees is called _____.

Select the correct option

☐	Bunch
☐	Cluster
☒	Forest
☐	Class

CORRECT ANS
MOHSIN CHINIOTI
03137248280

MOHSIN CHINIOTI

Question # 10 of 10 (Start time: 10:46:18 PM, 08 March 2022)

If a tree has 20 edges/links, then the total number of nodes in the tree will be :

▶ Select the correct option

☐	20
☐	Can't be determined
☐	21
☐	19

CORRECT ANS
MOHSIN CHINIOTI
03137248280

MOHSIN

Question # 2 of 10 (Start time: 09:47:12 PM, 08 March 2022)

A table consists of several columns, known as

Select the correct option

☐	Entity
☐	Fields
☐	Tuple
☐	Record

CORRECT ANS
MOHSIN CHINIOTI
03137248280

MOHSIN

Question # 3 of 10 (Start time: 09:48:20 PM, 08 March 2022)

Which of the following statement is NOT correct regarding Table ADT?

Select the correct option

☐	A major use of table is in databases where we build and use tables for keeping information.
☐	In a table, the type of information in columns may be different.
☐	The row of a table is called a record.
☐	A table consists of several columns, known as entities.

CORRECT ANS
MOHSIN CHINIOTI
03137248280

MOHSIN

Question # 4 of 10 (Start time: 09:49:57 PM, 08 March 2022)

Heap can be used to implement

Select the correct option

☐	Linked list
☐	Priority Queue
☐	Stack
☐	Queue

CORRECT ANS
MOHSIN CHINIOTI
03137248280

MOHSIN

Question # 5 of 10 (Start time: 09:51:34 PM, 08 March 2022)

During the union by size method, all the array element initialized to -1 shows:

Select the correct option

☐	Every tree has two elements each.
☐	Initial condition of tree
☐	None of given.
☐	Tree is complete now

CORRECT ANS
MOHSIN CHINIOTI
03137248280

REPEAT

MOHSIN

Question # 6 of 10 (Start time: 09:53:02 PM, 08 March 2022)

If the height of a perfect binary tree is 4. What will be the total number of nodes in it?

Select the correct option

☐	32	
☐	31	CORRECT ANS MOHSIN CHINIOTI 03137248280
☐	15	
☐	16	

MOHSIN

Question # 7 of 10 (Start time: 09:54:42 PM, 08 March 2022)

The array in binary search is sub divided _____.

Select the correct option

☐	Until a sublist is no more divisible	CORRECT ANS MOHSIN CHINIOTI 03137248280
☐	N time	
☐	Twice	
☐	Once	

MOH.

Question # 8 of 10 (Start time: 09:55:58 PM, 08 March 2022)

In 1990,Bill Pugh proposed an enhancement on linked lists and the new data structure was termed as

Select the correct option

☐	Enhanced Linked list
☐	Tree List
☐	Skip list
☐	Fixed list

CORRECT ANS
MOHSIN CHINIOTI
03137248280

MOHSIN

Question # 9 of 10 (Start time: 09:57:28 PM, 08 March 2022)

Which of the following properties are satisfied by Equivalence relationship?

Select the correct option

☐	Only Reflexive and Symmetric
☐	Only Reflexive and Transitive
☐	Reflexive , Symmetric and Transitive
☐	Only Symmetric and transitive

CORRECT ANS
MOHSIN CHINIOTI
03137248280

MOHSIN

Question # 10 of 10 (Start time: 09:58:35 PM, 08 March 2022)

If a tree has 50 nodes, then the total edges/links in the tree will be :

Select the correct option

☐	55
☐	50
☐	51
☐	49

CORRECT ANS
MOHSIN CHINIOTI
03137248280

MOHSIN

Question # 1 of 10 (Start time: 08:19:44 PM, 08 March 2022)

During union by size method, which data structure is used to improve the balancing of tree?

▶ Select the correct option

☐	Tree	
☐	Array	CORRECT ANS MOHSIN CHINIOTI 03137248280
☐	Stack	
☐	Linked List	

Question # 2 of 10 (Start time: 08:21:41 PM, 08 March 2022)

Which property of equivalence relation is satisfied if we say:
Ahmad R(is related to) Ahmad

▶ Select the correct option

☐	Reflexivity	CORRECT ANS MOHSIN CHINIOTI 03137248280
☐	All of the above	
☐	Transitivity	
☐	Symmetry	

Question # 3 of 10 (Start time: 08:23:26 PM, 08 March 2022)

Which of the following is NOT true regarding the maze generation?

Select the correct option

☐	Remove a randomly chosen wall if the cells it separates are already in the same set.
☐	Removing a wall is the same as doing a union operation.
☐	Randomly remove walls until the entrance and exit cells are in the same set.
☐	Do not remove a randomly chosen wall if the cells it separates are already in the same set.

CORRECT ANS
MOHSIN CHINIOTI
03137248280

MOHSIN

Question # 4 of 10 (Start time: 08:25:19 PM, 08 March 2022)

The expression
if (! heap->isEmpty())
checks

▶ Select the correct option

☐	Heap is full
☐	Heap is not empty
☐	Heap is empty
☐	Not a valid expression

CORRECT ANS
MOHSIN CHINIOTI
03137248280

MOHSIN

Question # 4 of 10 (Start time: 08:25:19 PM, 08 March 2022)

The expression
if (! heap->isEmpty())
checks

▶ Select the correct option

☐	Heap is full
☐	Heap is not empty
☐	Heap is empty
☐	Not a valid expression

CORRECT ANS
MOHSIN CHINIOTI
03137248280

MOHSIN

Question # 5 of 10 (Start time: 08:27:17 PM, 08 March 2022)

Which of the following properties are satisfied by Equivalence relationship?

Select the correct option

☐	Only Reflexive and Symmetric
☐	Reflexive , Symmetric and Transitive
☐	Only Reflexive and Transitive
☐	Only Symmetric and transitive

CORRECT ANS
MOHSIN CHINIOTI
03137248280

MOHSIN

Question # 6 of 10 (Start time: 08:28:52 PM, 08 March 2022)

For a perfect binary tree of height h , having N nodes, the sum of heights of nodes is _____.

▶ Select the correct option

☐	$N - h - 1$	
☐	$N - 1 + h$	CORRECT ANS MOHSIN CHINIOTI 03137248280
☐	$N - (h - 1)$	
☐	$N - 1$	

MOHSIN

Question # 6 of 10 (Start time: 08:28:52 PM, 08 March 2022)

For a perfect binary tree of height h , having N nodes, the sum of heights of nodes is _____.

▶ Select the correct option

☐	$N - h - 1$
☐	$N - 1 + h$
☐	$N - (h - 1)$
☐	$N - 1$

CORRECT ANS
MOHSIN CHINIOTI
03137248280

MOHSIN

Question # 7 of 10 (Start time: 08:30:43 PM, 08 March 2022)

There are 100 elements in a heap, if we perform 100 deleteMin operations then we will get _____ list

▶ Select the correct option

☐	Sorted
☐	Unsorted
☐	Random
☐	Nonlinear

CORRECT ANS
MOHSIN CHINIOTI
03137248280

MOHSIN

Question # 8 of 10 (Start time: 08:33:03 PM, 08 March 2022)

If there are 100 elements in an equivalence class then we will have _____ sets initially.

Select the correct option

☐	1000
☐	80
☐	100
☐	50

CORRECT ANS
MOHSIN CHINIOTI
03137248280

MOHSIN

Question # 9 of 10 (Start time: 08:34:56 PM, 08 March 2022)

Binary Search can be categorized into which of the following?

Select the correct option

☐	Divide and conquer	CORRECT ANS MOHSIN CHINIOTI 03137248280
☐	Greedy algorithm	
☐	Brute Force technique	
☐	Dynamic programming	

MOHSIN

Question # 10 of 10 (Start time: 08:36:31 PM, 08 March 2022)

The array in binary search is sub divided _____.

Select the correct option

☐	Once
☐	N time
☐	Until a sublist is no more divisible
☐	Twice

CORRECT ANS
MOHSIN CHINIOTI
03137248280

MOHSIN

A TEAM BY VU SQUAD

CS-301 FINAL TERM NOTICE PREPARED BY
MOHSIN CHINIOTI

VUSQUAD134@GMAIL.COM

[03137248280](tel:03137248280)

Q 1: Name of two divide and conquer algorithm?

Merge Sort

Quick Sort

Heap Sort

These Three algorithm fall under "Divide and Conquer" Category.

Q 2: Difference between call by value and call by reference?

In case of call by value, a copy of object is made and placed at the time of function calling and By using the references, we are not making the copy but send the address of the variable.

Q 3: Define heap and two types of heap?

It is used in priority queue. The definition of heap is that it is a complete binary tree that conforms to the heap order.

Here heap order means the min and max heap. In max heap, each node has a value greater than the value of its left and right child nodes.

In Min heap the value in the parent node is less than the value on its children nodes.

Q 4: Height of a tree is 5 find the sum of heights?

Height = 5

Nodes = $2^h = 2^5 = 32$

Sum of Nodes = $N - (h + 1) = 32 - (5 + 1) = 32 - 6 = 26$

Q 5: Write one example of hashing?

Hashing can be used to store the data. Suppose we have a list of fruits. The names of Fruits are in string. The key is the name of the fruit. We will pass it to the hash Function to get the hash key.

Hash Code ("apple") = 5 we pass it to the string "apple". Resultantly, it returns a number 5.

Q 6: How do we carry out degeneration of Complete Binary Tree?

Degeneration of the (original) binary tree is mainly reflected by degeneration of the internal infix trees and less by degeneration of the external height of the binary tree.

Q 7: What is a skip List?

A skip list for a set S of different items is a series of lists $S_0, S_1, \dots, S_h$. S_0 ,

Each list S_i contains the special keys $+\infty$ and $-\infty$

List S_0 contains the keys of S in non-decreasing order

Each list is a result of the previous one,

List S_h contains only the two special keys

Q 8: How we can generate a maze with the help of union?

Answer:

The algorithm is as follows:

Randomly remove walls until the entrance and exit cells are in the same set.

Removal of the wall is the same as doing a union operation.

Do not remove a randomly chosen wall if the cells are already in the same set.

Q 9: Write down the C++ code to implement insertion sort algorithm.

Answer:

Following is the code of the insertion sort in C++.

```
Void insertionSort(int *arr, int N) { int pos,
count, val;
```

```
For (count=1; count < N; count++)
```

```
{
```

```
val = arr[count];
```

```
For (pos=count-1; pos >= 0; pos--)
```

```
If (arr[pos] > val)
```

```
Arr [pos+1] =arr [pos];
```

```
Else break;
```

```
Arr [pos+1] = val;}}
```

Q 10: What is Table ADT? Discuss any two implementations of table ADT.

Answer:

The table, an abstract data type, is a collection of rows and columns of information.

Implementation of Table

1. Sorted Sequential Array
2. Unsorted Sequential Array

Q 11: Suppose we have the following representation for a complete Binary Search Tree, tell the Left and Right child nodes and Parent node of the node D.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	...
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	...

Answer:

Parent = B

By using formula of parent ($i/2$)

As "i" is equal to array element D which place on size 4, so $i/2 = 4/2 = 2$ as Array element B is on size 2 So its Parent is "B" according to the formula.

Left child = H

By using formula of Left Child $2i$

So, $2i = 2 \times 4 = 8$

As Array element H place on size 8 so it's left child is "H" according to formula.

Right child = I

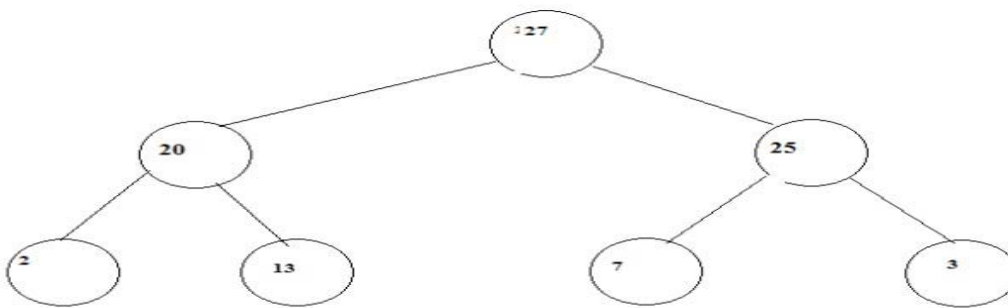
By using formula of Right Child $2i+1$

So, $2i + 1 = (2 \times 4) + 1 = 9$

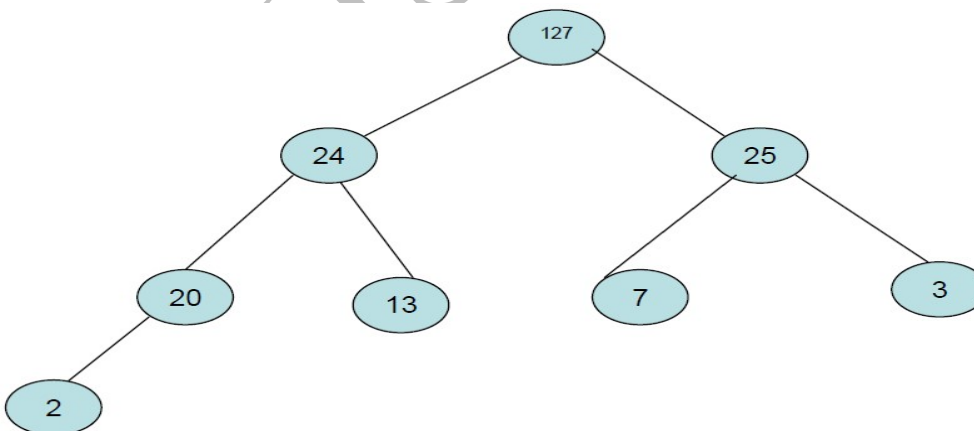
As Array element I place on size 9 so its Right Child is "I" according to formula.

P-2

Q 12: Consider the following max heap. Add node 24 in it and show the resultant Heap.



Answer:



Q 13: Here is an array of 10 integers. 5, 3, 8, 9, 1, 7, 0, 2, 6, 4, Draw this array after the first iteration of the large loop in a selection sort (Sorting from smallest to largest) Answer:

P-3

	5	3	8	9	1	7	0	2	6	4
1 st step	0	3	8	9	1	7	5	2	6	4
2 nd step	0	1	8	9	3	7	5	2	6	4
3 rd step	0	1	2	9	3	7	5	8	6	4
4 th step	0	1	2	3	9	7	5	8	6	4
5 th step	0	1	2	3	4	7	5	8	6	9
6 th step	0	1	2	3	4	5	7	8	6	9
7 th step	0	1	2	3	4	5	6	8	7	9
8 th step	0	1	2	3	4	5	6	7	8	9

Q 14: What is an Equivalent relation or Class? Give any two examples.

A binary relation R over a set S is called an *equivalence relation* if it has following properties. Reflexivity

Symmetry

Transitivity

Example; sets of peoples, electric circuit domain

Q 15: How heap sort works to set a set of data.

Min heap and max heap are used to sort the data. Min heap convert the data in ascending sorted order and max heap convert the in descending order.

Q 16: How can we search an element in skip list?

We search for a key x in the following way, We start at the first position of the top list. At the current position p, we compare x with y. $x = y$: we return element. $x > y$: we scan forward. $x < y$: we drop down.

Q 17: What is an ADT?

The table, an abstract data type, is a collection of rows and columns of information.

Q 18: How we can implement Table ADT using linked?

Linked list is one choice to implement table abstract data type. For unsorted elements, the insertion at *front* operation will take constant time. But if the data inside the list is kept in sorted order then to insert a new element in the list, the entire linked list is traversed to find the appropriate position for the new element.

Q 19: If we allow assignment to constants what will happen?

Suppose "a" has the value number 3. Now we assigned number 2 the number 3 i.e. all the number 2 will become number 3 and the result of $2 + 2$ will become 6. Therefore it is not allowed.

Q 20: Explain the process of Deletion in a Min-Heap.

We have to write a function *deleteMin()*, which will find and delete the minimum number from the tree. Finding the minimum is easy; it is at the top of the heap.

Q 21: Give any three characteristics of Union by Weight method.

Following are the most important characteristics of this method:

Maintain sizes (number of nodes) of all trees, and during *union*.

Make smaller tree,

Implementation: for each root node i , instead of setting $\text{parent}[i]$ to -1 , set it to $-k$ if tree rooted at i has k nodes. This is also called *union-by-weight*.

Q 22: "For smaller lists, linear insertion sort performs well, but for larger lists, quick sort is suitable to apply." Justify why?

For smaller lists number of comparisons and interchange decreases fundamentally in insertion sort than quick sort. For larger lists, opposite occurs in both sorting techniques.

Q 23: What is Disjoint Sets? Explain with an example.

Two sets are disjoint if they have no elements in common.

Example:-Suppose there are many people around you. We make groups of people who are related to each other. The people in one group will say that they are related to each other. Similarly we can have many groups. The people in the group will say that there is not a single person in the other group who is related to them.

P-4

Q 24: Give the difference between strict and complete binary tree.

A binary tree said to be a strictly binary tree if every non-leaf node in a binary has non-empty left and right sub trees. If there are left and right sub trees for each node in the binary tree show a complete binary tree.

Q 25: A complete binary tree can be stored in an array. While storing the tree in an array we leave the first position (0th index) of the array empty. Why?

In case of having a node with left and right children, stored at position i in the array, the left child will be at position $2i$ and the right child will be at $2i+1$ position. If the value of i is 2, the parent will be at position 2 and the left child will be at position $2i$ i.e. 4. The right child will be at position $2i+1$ i.e. 5. We have not started the 0th position. It is simply due to the fact that if the position is 0, $2i$ will also become 0. So we will start from the 1st position, ignoring the 0th.

Q 26: Give the effect of sorted data on Binary Search.

The binary tree is a fundamental data structure used in computer science. The binary tree is a useful data structure for rapidly storing sorted data and rapidly recovers stored data.

Q 27: Give your comment on the statement that heap uses least memory in array representation of binary trees. Justify your answer in either case.

A heap is a complete binary tree, so it is easy to be implemented using an array representation it is a binary tree with the following properties:

Property 1: it is a complete binary tree.

Property 2: the value stored at a node is greater or equal to the values stored at the children.

Q 28: Explain the following terms:

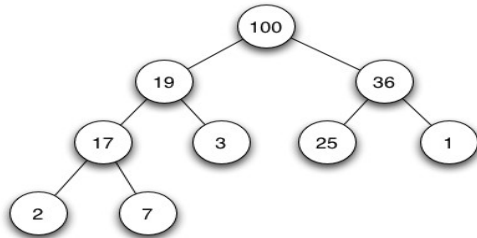
1. Collision

2. Linear Probing 3. Quadratic Probing Answer:

Collision: When two values confusion to the same array location, this is called a *collision*. **Linear Probing** When there is a collision, some other location in the array is found. This is known as linear probing.

Quadratic Probing In the quadratic probing when a collision happens we try to find the empty location at index $+ 1^2$. If it is filled then we add 2^2 and so on.

Q 29: Convert this tree representation of a heap into the corresponding array representation.



Answer:

100	19	36	17	3	25	1	2	7
-----	----	----	----	---	----	---	---	---

Q 30: What are different applications of Hashing?

Compilers use hash tables to keep track of declared variables.

1. A hash table can be used for on-line spelling checkers. If misspelling detection (rather than correction) is important, an entire dictionary can be hashed and words checked in constant time.
2. Game playing programs use hash tables to store seen positions,
3. Hash functions can be used to quickly check for inequality. If two elements hash to different values they must be different.

Q 31: Give the operation names that we can perform on Table abstract data type.

Insert
Find
Remove

Q 32: Give your comments on the statement: "Efficiently developed data structures decrease programming effort"

Reduces programming effort by providing useful data structures and algorithms so you don't have to write them yourself,

Increases performance by providing high-performance implementations of useful data structures and algorithms,

Q 33: When Hashing is NOT suitable?

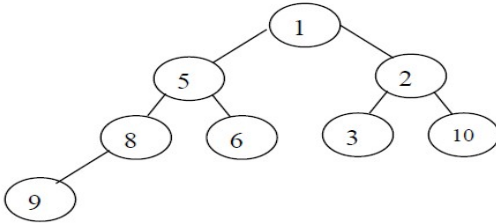
Hash tables are not so good if there are many insertions and deletions, or if table traversals are needed in this case, AVL trees are better.

Q 34: Heapify the elements of the following array (reading from left to right) into a Min Heap and show that Min Heap contents in the form of array as shown below,

original array	6	5	3	9	1	2	10	8	-
----------------	---	---	---	---	---	---	----	---	---

Answer:

Min heap



Heapified array 1 5 2 8 6 3 10 9

Q 35: Here is an array of ten integers: 5 3 8 9 1 7 0 2 6 4 Show the first three merging steps for Merge sort on this array.

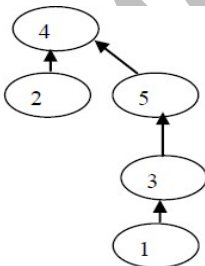
P-6

	5	3	8	9	1	7	0	2	6	4
1 st step	5	3	8	9	1	7	0	2	6	4
2 nd step	3	5	8	9	1	7	0	2	4	6
3 rd step	3	5	8	9	1	7	0	2	4	6
4 th step	1	3	8	5	9	2	6	7	0	4
5 th step	1	3	8	5	9	2	6	7	0	4
6 th step	1	2	3	6	7	8	0	5	4	9

Q 36: Consider the following sequence of union commands on the set of elements {1, 2, 3, 4, 5}:



Union (4, 2) union (3, 1) union (5, 4) union (5, 3) Show the result when the unions are performed.
Answer:



Q 37: Let's call the node as a that requires re-balancing. Consider two cases below: 1)
An insertion into right sub tree of the left child of a.

2) An insertion into left sub tree of the right child of a.

Which of the following statement is correct about the above two cases.

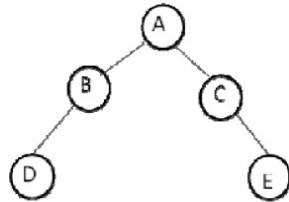
- 1) The insertion occurs outside (i.e., right to left or left to right) in the above two cases, single rotation can fix the balance in these two cases.
- 2) The insertion occurs inside ((i.e., right to left or left to right) in these two cases, single rotation cannot fix the balance in these two cases.

Answer:

Statement 2 is correct about the above two cases.

The insertion occurs inside ((i.e., right to left or left to right) in these two cases, single rotation cannot fix the balance in these two cases.

Q 38: Represent the following Binary tree using array representation.

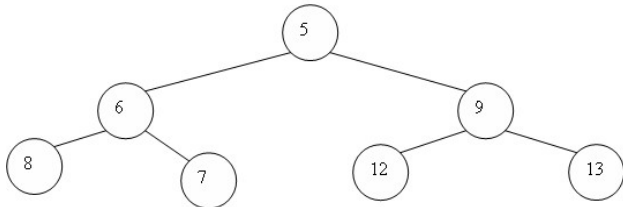


Answer:

A	B	C	D	E
---	---	---	---	---

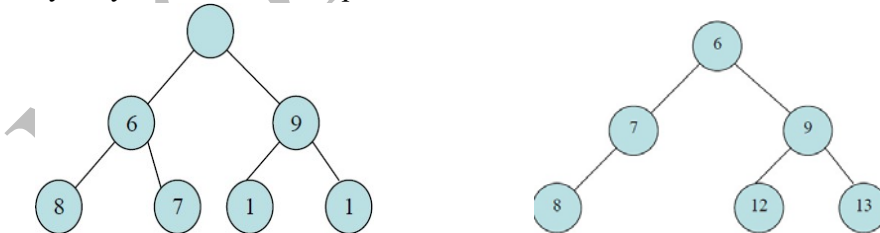
Q 39: How we can apply Find operation on elements combined through Union operation. We give an element to the *find* method and it returns the name of the set.

Q 40: Consider the following Min Heap apply operation delMin on it and show the resultant Heap,



Answer:

It is clear from the definition of min-heap that it should be in the *root* of the tree. So finding the minimum is very easy. Here node 5 is operational node that has to be deleted.



Q 41: Which implementation of disjoint set is better with respect space and why?

An array that is a single dimensional structure and seems better with respect to space, Suppose we have 1000 set members i.e. names of people. If we make a Boolean matrix for 1000 items, its size will be 1000×1000 . Thus we need 1000000 locations for Boolean items; its size will be

1000 × 1000. Thus we need 1000000 locations for Boolean values. But in case of an array the number of location will be 1000. As array is single dimensional structure.

Q 42: Suppose the following numbers are stored in an array „BS“: 32, 51, 27, 85, 66, 23, 13, 57 We apply Bubble sort to array „BS“. Show step by step sorting. You have to write only two passes that is passing 1 and pass 2.

Hint; while concern step by step sorting there are 6 passes involved to sort array num. You have to show all passes **Answer:**

Pass one:-

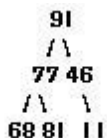
	32	51	27	85	66	23	13	57
Step1	32	51	27	85	66	23	13	57
Step2	32	27	51	85	66	23	13	57
Step3	32	27	51	66	85	23	13	57
Step4	32	27	51	66	23	85	13	57
Step5	32	27	51	66	23	13	85	57
Step6	32	27	51	66	23	13	57	85

Pass two:-

	32	27	51	66	23	13	57	85
Step1	32	27	51	66	23	13	57	85
Step2	27	32	51	66	23	13	57	85
Step3	32	27	66	23	66	13	57	85
Step4	32	27	66	23	13	66	57	85
Step5	32	27	66	23	13	57	66	85
Step6	32	27	66	23	13	57	66	85

P-8

Q 43: Give two different reasons to explain why the following binary tree is not a heap:



Answer:

Above array is not min heap or max heap because every parent node has not minimum or maximum value. In max heap, each parent node has a value greater than the value of its left and right child nodes. In min heap, each parent node has a value less than the value of its left and right child nodes.

Q 44: Describe the conditions for second case of deletion in AVL Trees.

This is the case where the parent of the deleted node had a balance of 1 and the node was deleted in the parent's left sub tree.

Q 45: How many leaf and non-leaf nodes are present in a complete binary tree if its depth is 7?

Leaf Node = $2^d = 2^7 = 128$

Non leaf Nodes = $2^d - 1 = (2^7) - 1 = 128 - 1 = 127$

Q 46: Write C++ algorithm for Binary Search. int

isPresent(int *arr, int val, int N)

{ int low = 0;

int high = N - 1;

int mid;

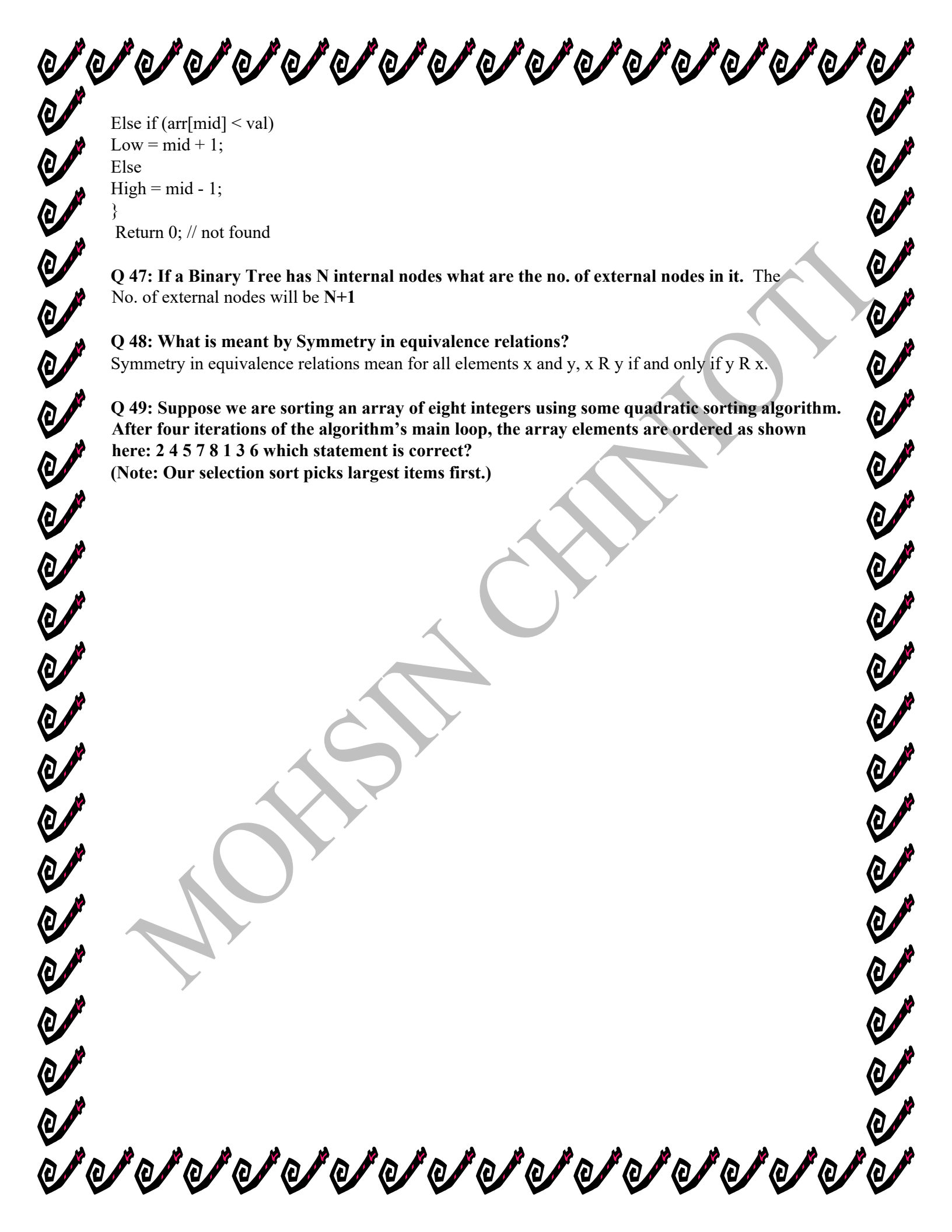
While (low <= high)

{

Mid = (low + high)/2; If

(arr[mid] == val)

Return 1; // found!



```
Else if (arr[mid] < val)
Low = mid + 1;
Else
High = mid - 1;
}
Return 0; // not found
```

Q 47: If a Binary Tree has N internal nodes what are the no. of external nodes in it. The No. of external nodes will be N+1

Q 48: What is meant by Symmetry in equivalence relations?
Symmetry in equivalence relations mean for all elements x and y, $x R y$ if and only if $y R x$.

Q 49: Suppose we are sorting an array of eight integers using some quadratic sorting algorithm. After four iterations of the algorithm's main loop, the array elements are ordered as shown here: 2 4 5 7 8 1 3 6 which statement is correct?
(Note: Our selection sort picks largest items first.)

- A. The algorithm might be either selection sort or insertion sort.
B. the algorithm might be selection sort, but it is not insertion sort.
C. the algorithm is not selection sort, but it might be insertion sort.
D. The algorithm is neither selection sort nor insertion sort.
E. None of these. Answer:

The correct statement is,

C. The algorithm is not selection sort, but it might be insertion sort.

Q 50: If we insert a new element into an AVL tree of height 4, is one rotation sufficient to reestablish balance? Justify your answer.

No, one rotation is not always sufficient to re-establish balance.

Q 51: Write down 3 important characteristics of skip list.

1. Fast Searching of Sorted Chain.
2. Provide alternative to BST (binary search trees) and related tree structures.
3. Search and update require linear time.

Q 52: Drawback of using array to store BST.

The drawback is the limitations in which array store fixed number of elements. Therefore, no more elements can be stored after fixing the array size.

Q 53: C++ code to add node in doubly link list. Class

Node { Public: int get() { return object; }; // returns the value of the element

Void set (int object) {this->object = object ;}; // set the value of the element

Node* getNext() { return nextNode; }; // get the address of the next node

Void setNext(Node* nextNode) // set the address of the next node

{This->nextNode = nextNode;};

Node* getPrev() { return prevNode; }; // get the address of the prev node

Void setPrev(Node* prevNode) // set the address of the prev node

{This->prevNode = prevNode;}; Private: int

object; // it stores the actual value of the element

Node* nextNode; // this points to the next node

Node* prevNode; // this points to the previous node

};

Q 54: What you conclude from the running time analysis of disjoint sets.

We conclude that:

1. Union is clearly a constant time operation.
2. Running time of find (i) is proportional to the height of the tree containing Node i.
3. This can be proportional to n in the worst case (but not always)

Q 55: what is C++ template?

A template is a function or class that is written with a generic data type. When a programmer uses this function or class the generic data type is replaced with the data type needed to be used in the template function or in the template class. The compiler automatically creates a version of that function or class with that specified data type.

Q 56: Explain and write the code for union and find operation of parent array in disjoint sets.

Answer:

Find (i)

This loop is used to find the parent of an element.

// traverse to the root (-1)

For (j=i; parent[j] >= 0; j=parent[j])

Return j;

Union (i, j) Now let's see the code for the function of union. Here we pass two elements to the function union. The union finds the roots of i and j.

Root i = find (i);

Root j = find (j);

If (root i! = root j)

Parent [root j] = root i;

Q 57: Convert the given infix form to postfix form.

A+B/C-D^E-F Answer:

AB+C/D-EF -^

Q 58: Write the code of the percolateDown () function and also comment it.

Template <class eType>

Void Heap<eType>::percolateDown(int hole)

{ int

child;

eType tmp = array[hole];

For (hole * 2 <= currentSize; hole = child)

{Child = hole * 2;

If (child! = currentSize && array [child+1] < array [child])

Child++; *// right child is smaller*

If (array [child] < tmp)

Array [hole] = array [child];

Else break ;}

Array [hole] = tmp;}

Q 59: Which three basic operations are performed on AVL TREES?

(1) Insert (2) Delete (3) Search

Q 60: Write the applications of heap.

The use of priority queue with the help of heap is a major application. The priority queue is itself a major data structure, much-used in operating systems. Similarly Heap data structure is also used in sorting algorithms.

Q 61: Write the application of binary search tree.

The searching and sorting operations are very common in computer science with the help of binary search tree.

Q 62: Write the any two operation of linked list that perform their function in one step.

(1) Add (2) Remove (3) Find

Q 63: Write different type of hashing.

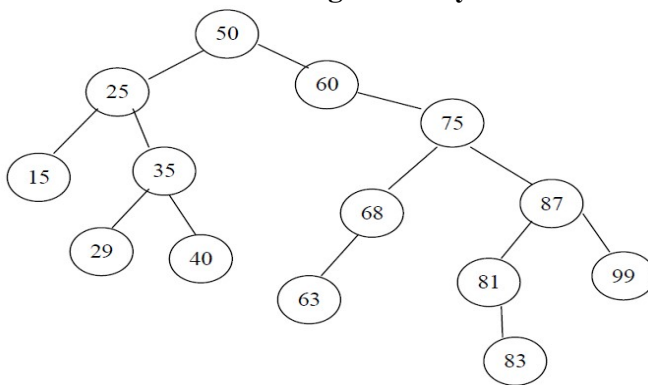
Data structures that employ hashing methods to store keys are denoted as **hashing types**.

Q 64: What is Dynamic Equivalence Problem with an example?

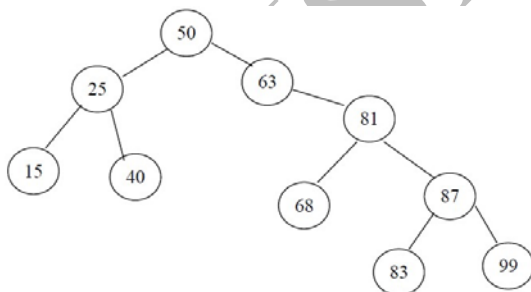
Suppose we have a set and converted into disjoint sets. Now we have to decide if these disjoint sets hold equivalence relation or not.

For example of family relations in which we had divided the people into groups depending upon their relations.

Q 65: Consider the following Binary tree and Delete the key 29, 35, 75 and 60 only. Show the resultant tree after deleting these keys.



Answer:



Q 66: Two ways for the implementation of Queue data structure.

1. Implementation of queue using linked list.
2. Implementation of queue using Array.

Q 67: Worst case of selection sort.

The worst case occurs if the array is already sorted in descending order.

P-12

Q 68: “Is a sibling of” set of all human beings is equivalence relation or not? Explain. Yes

it is an equivalence relation,

The transitivity is: _for all elements x, y and z . If $x R y$ and $y R z$, then $x R z$. suppose x is Harris, y is Saad and z is Ahmed. If Harris is related to Saad, Saad is related to Ahmed. We can deduce that Harris is related to Ahmed. This is also true in relationships. If you are cousin of someone, the cousin of that person is also related to you.

Q 69: What is Linked Memory?

Linked memory is a memory in which the various cells of memory are not located continuously. In this process, each cell of the memory not only contains the value of the element but also the information where the next element of the list is exist in the memory.

Q 70: Code for removing an element from a linked list data structure if it is implementation by using linked memory methods.

```
Void remove () {  
If (current Node! = NULL && current Node! = head Node) {  
LastCurrentNode->set Next (current Node->getNext ());  
Delete current Node;  
Current Node = lastCurrentNode->getNext ();  
Size--;  
} };
```

Q 71: What is table abstract data type?

Table abstract data type is used for the method of insertion, removing and finding. Linked list is one choice to implement table abstract data type. For unsorted elements, the insertion at *front* operation will take constant time.

Q 72: In the array representation of union what represents -1? This
shows that node has no parent.

Q 73: If we want to delete the node from BST which has left and right child then which rotation is applied?

After deleting the node we traverse up the tree from the deleted node to checking the balance of each node at each level up of the root.

Q 74: Write Algorithm union by weight?

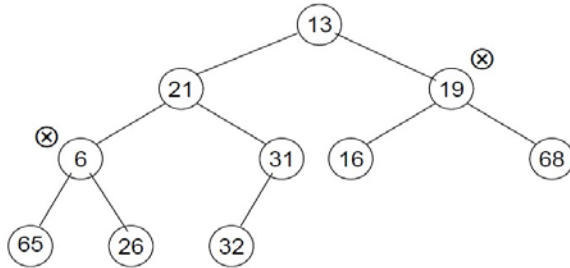
```
//union (i, j):  
root1 = find (i);  
root2 = find (j); If  
(root1! = root2)  
If (parent [root1] <= parent [root2]) {  
// first tree has more nodes  
Parent [root1] += parent [root2];
```

```

Parent [root2] = root1 ;}
Else {
// second tree has more nodes
Parent [root2] += parent [root1];
Parent [root1] = root2 ;}

```

Q 75: A tree is given, Tell is it heap or not if it is heap then write its type?



Answer:

Look at the node having value 19. The values in its left and right child nodes are 16 and 68 respectively. Thus the value of left child (i.e. 16) is less than that of the parent. So it is not a heap. If it were a heap or min heap, the value 16 should have been parent and the value 19 should have its child. Due to this violation, (the value of child is less than that of the parent) it is not a heap.

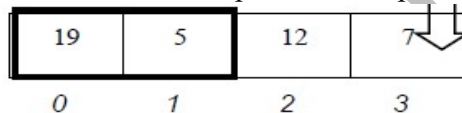
Q 76: Which data structure is best for priority queue?

The priority queue is itself a major data structure, much-used in operating systems.

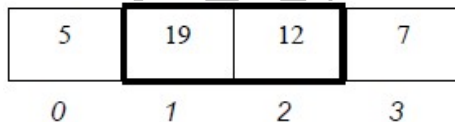
Q 77: One array is giving require sorting it using bubble sort and write only 2 iterations?

Array that has elements 19, 12, 5 and 7 Answer:

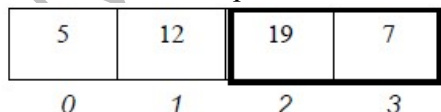
First of all, we compare the first pair i.e. 19 and 5. As 5 is less than 19, we exchange these elements.



Now 5 is at its place and we will take the next pair. This pair is 19, 12 and not 12, 7. In this pair 12 is less than 19, we exchange 12 and 19.

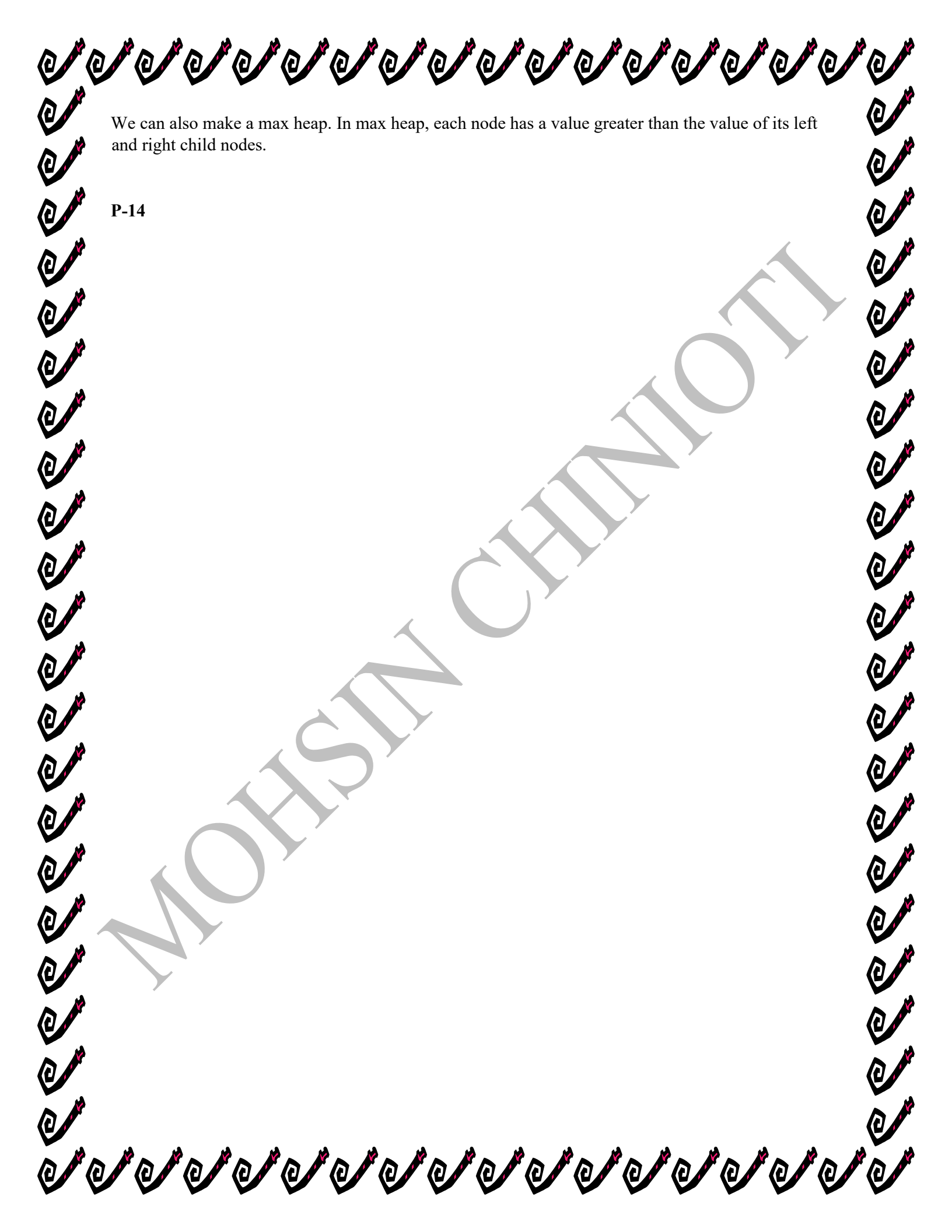


After this, the next pair is 19, 7. Here 7 is less than 19 so we exchange it.



Now 7 is at its place as compared to 19 but it is not at its final position. The element 19 is at its final position. Now we repeat the pair wise exchange on the array from index 0 to 2 respectively.

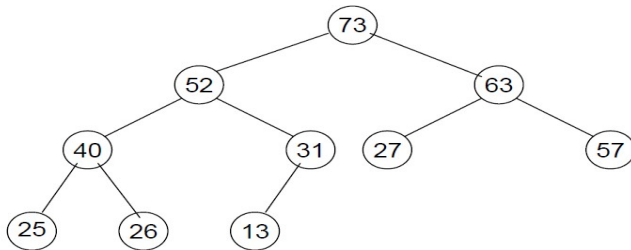
Q 78: Define Max Heap?



We can also make a max heap. In max heap, each node has a value greater than the value of its left and right child nodes.

P-14

MOHSIN CHINNOTI



Max Heap

Q 79: Give the effect of sorted data on Binary Search. It gives a look of a linked list.

Q 80: Suppose the hash table of height 7 (index 0 to 6), hash function $H(\text{key}) = (2 * \text{key} + 5) \text{ Mode } 7$ and pass these numbers [5, 23, 17, 14, 44] from the hash function to resolve the collision by the linear probing.

As we know that, $H(\text{key}) = (2 * \text{key} + 5) \text{ Mode } 7$

So,

$$H(5) = (2 * 5 + 5) \text{ mode } 7 = 2 * 5 + 5 = 15 \text{ mode } 7 = 1$$

1 is reminder after dividing 15 by 7. And this reminder is Mode.

$$H(23) = (2 * 23 + 5) \text{ mode } 7 = 2$$

2 is reminder after dividing 51 by 7. And this reminder is Mode.

$$H(17) = (2 * 17 + 5) \text{ Mode } 7 = 4$$

4 is reminder after dividing 39 by 7. And this reminder is Mod. And

So on...

Q 81: Why binary search not use with link list?

Because it store sorted data,

Q 82: What is the use of *this* pointer?

This pointer is used to point out the current object. We need this pointer to add or remove current node from the list.

Q 83: Array is given

3 5 9 8 1 7 0 2 6 4

Draw after 1st iteration of main loop in selection sort algorithm. 0

5 9 8 1 7 3 2 6 4

Q 84: Sort the data items given below with insertion sort algorithm. Write each step 77

33 44 88 11 22 66 55

33 77 44 88 11 22 66 55

33 44 77 88 11 22 66 55

33 44 77 88 11 22 66 55

33 44 11 77 88 22 66 55

11 33 44 77 88 22 66 55

11 33 44 77 22 88 66 55

11 33 44 22 77 88 66 55

...

11 22 33 44 55 66 77 88 = final

P-15

Q 85: hash function was given and $f(h) = \text{key} \bmod 12$ identify key that may cause collusion.

36, 30, 84 Answer:

$$F(h) = 36 \bmod 12 = 12 * 3 + 0 = 0$$

$$F(h) = 30 \bmod 12 = 12 * 2 + 6 = 6$$

$$F(h) = 84 \bmod 12 = 12 * 7 + 0 = 0$$

So key 36 and 84 collusion each other.

Q 86: Properties of good hash function?

If the keys are integers, $\text{key} \% T$ is generally a good hash function, For example, if $T = 10$ and all keys end in zeros, then $\text{key} \% T = 0$ for all keys.

Q 87: What is image segmentation?

In image segmentation, we will divide the image into different parts. An image may be segmented according to the intensity of the pixels.

Q 88: In which traversal method recursion is not allowed?

In post order traversal method

Q 89: AVL insertion in which single rotation is successful?

An insertion into left sub tree of the left child of α

An insertion into right sub tree of the right child of α

Q 90: How one can find the height of a binary tree?

The highest number of nodes t starting from the first node (ROOT) to a leaf node is called the height of tree. The formula for finding the height of a tree $h = l(\text{max}) + 1$, where h is the height and l is the max level of the tree, we can say that height of tree can be calculated as maximum level of the tree.

Q 91: Why do we use Huffman encoding?

We use Huffman encoding to compress the size of a text document. Huffman code is also part of the JPEG image compression scheme.

Q 92: How we can use disjoint sets for image segmentations.

For the image segmentation, disjoint sets and *union-find* algorithm are very useful. In the image analysis course, you will actually apply these on the images.

Q 93: What is the first step of Huffman encoding algorithm.

In the first step, we make a list of all letters (characters) including space and end line character and find out the number of occurrences of each letter/character.

Q 94: Create frequency table from "Huffman Encoding". There is a text string as written below.

Traversing threaded binary trees

The size of this character string is 33 (it includes 3 space characters and one new line character). In the first step, we perform the counting of different characters in the string manually. We do not

assign a fake or zero frequency to a letter that is not present in the string. A programmer may be concerned only with the characters/letters that are present in the text.

P-16

Character	frequency	character	frequency
NL	1	I	2
SP	3	n	2
A	3	r	5
B	1	s	2
D	2	t	3
E	5	v	3
G	1	y	1
H	1		

Table 1: Frequency table

Q 95: The inclusion criteria of pixel?

An image is a collection of pixels. We want to separation the pixels into disjoint sets, one set per blob.

Q 96: Define Heap sort?

Let's say we have a large unsorted list of names and want to sort it. This list can be sorted with the help of the heap sort algorithm.

Q 97: Define Merge sort or bubble sort?

Merge sort works with arrays as well as linked lists. The merge sort algorithm is called $n \log 2n$ algorithm.

The third sorting algorithm is bubble sort. The basic idea of this algorithm is that we bring the smaller elements upward in the array step by step and as a result, the larger elements go downward.

Q 98: which process places data at the back of the queue?

Enqueue(X) Place X at the *rear* of the queue

Q 99: If data is in sorted order what will be its impact on find operation of table ADT? The find method is given a key and it finds the entry associated with the key. For example, in employees table if *employee id* is the key, we can find the record of an employee whose *employee id* is, say, 15466.

Q 100: how many parameters are required for performing the following operations on table also tell name of these parameters.

Insert: this method is used to insert (add) a record in a table.

Find: The find method is given a *key* and it finds the entry associated with the key.

Remove: Then there is the remove method that is given a value of the *key* to find and remove the entry associated with that key from the table.

Q 101: Write the remove () function Code in the Queue.

```

Void remove () {
If (currentNode != NULL && currentNode != headNode) {
(Step 1) lastCurrentNode->setNext (currentNode->getNext ());
(Step 2) delete currentNode;
(Step 3) currentNode = lastCurrentNode;
(Step 4) size--;
}
};

```

P-17

Q 102: Write usage of Table ADT.

A major use of table is in Databases where we build and use tables for keeping information.

Q 103: Define length () method in Queue.

Length () Returns the length of the list. The function length () tells us about the number of elements in the list.

// returns the size of the list

```

int length() {
Return size;
};

```

Q 104: Write Advantage of using Heap Sort.

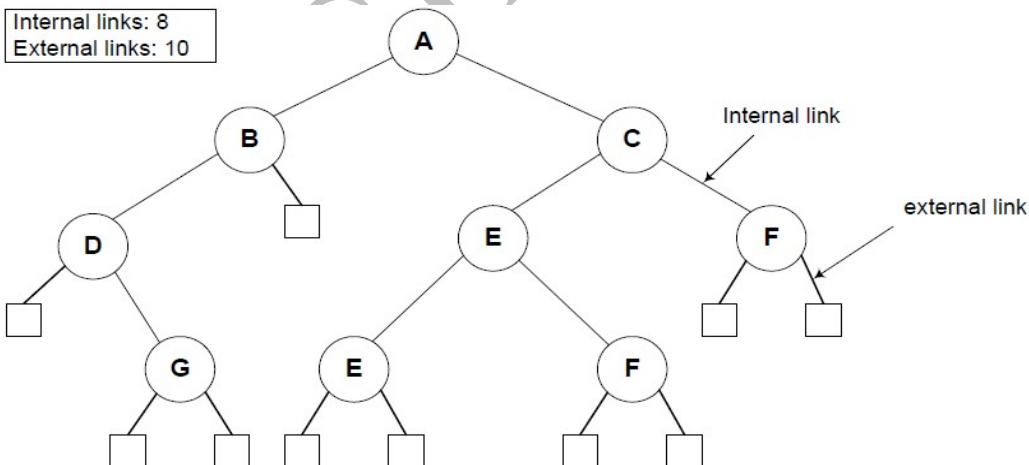
List can be sorted with the help of the heap sort algorithm.

Q 105: write one use of heap?

The major use of heap is in priority queues. Priority queues are not FIFO or LIFO.

Q 106: Find the internal link (N-1), external links (N+1), N =23

Internal links: 8
External links: 10



A binary tree with N internal nodes has 2N links, N-1 links to internal nodes and N+1 links to external nodes. Thus $N-1+N+1=2N$ links. And we will solve this by value of $N = 23$ given.

Q 107: Write the C++ code which will print the integers from 1 to 100 through the recursive function. #include<iostream>
Using namespace std;

```
Class A
{ Public:
    Static int a;
    A ()
    {Cout<<a++<<endl ;}
};
```

P-18

Int A::a = 1;

```
Intmain ()
{
    Int N = 100;
    Aobj[N];
    Return 0;
}
```

Q 108: Write public method of remove first () into the linked list, that remove the 1st item of the list.

```
Public void remove First () {
If (head == null)
Return;
Else {
If (head == tail) {
Head = null;
Tail = null;
} else {
Head = head. Next;
}}}
```

Q 109: How many rotations will be required to delete a node in AVL tree? If the delete node balance 0 and the node was deleted in the parents.

- 1) Left sub tree
- 2) Right sub tree

The first case is that the parent of the deleted node had a balance of 0 and the node was deleted in the parent's left sub tree. When we delete a node from its left sub tree then height of its right sub tree becomes larger than the left sub tree. Now the action that will be taken in this case is that, change the balance of the parent node and stop. No further effect on balance of any higher node. There is no need of rotation in this case. Thus it is the easiest case of deletion.

And in second case the parent of the deleted node had a balance of 0 and the node was deleted in the parent's right sub tree. After removing the right child the balance of it changes and it becomes 1, as the height of its left sub tree is 1 greater than the height of right sub tree. The action performed in this

case is the same as that in case 1. That is change the balance of the parent node and stop. No further effect on balance of any higher node.

Q 110: Suppose the following function is define in a class named “A” Int

```
(inty) const
{
    Coutyendl;
    Intx;
    X=y+10; Coutx
};
```

what is the output of the function when 20 is passed to y???

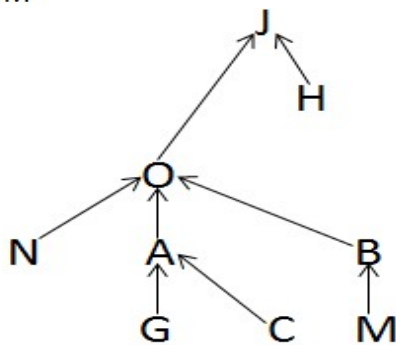
P-19

As given that Intf (inty) const. This means Y is constant variable and we know that the value of constant variable can't change; it will give compile time error, as constant variable cannot be change, so the outputs of function will y + 10. We cannot pass 20 to “y”.

Q 111: Construct the tree using union field data structure. Relations are given bellow:

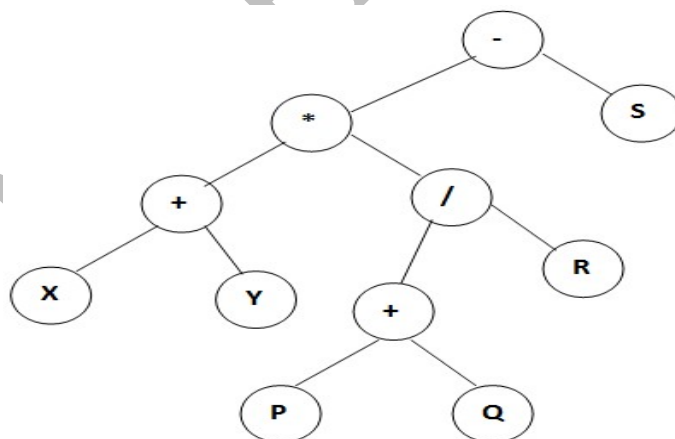
Union(O,A),union(B,M),union(A,G),union(O,N),union(J,H),union(A,C),union(O,B)union(J,O)

O B J
↑ ↑ ↑
A M H

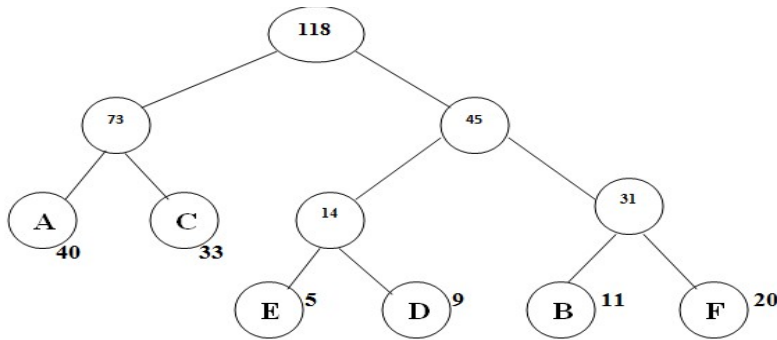


Q 112: Draw expression tree for the following arithmetic expression:

$((x + y) * (p + q)/r) - S$



Q 113: Making Huffman binary tree using A, b, c, d, e, f and their frequency is 40, 11,33,9,5,20.



P-20

Q 114: Evaluate the following postfix expression.

6165-+*1121///

According to rule of postfix expression to infix, the operator will shift in between last two operands.

So,

6165-+*	1121///
61(6-5)	11(2/1)
6(1+ (6-5))	1(1/ (2/1))
(6*(1+ (6-5)))	(1/ (1/ (2/1)))

Q 115: Suppose the equivalent relation is defined over the five elements {a1, a2, a3, a4, a5} the elements can be name of people, pixels, electrical component etc. how many pairs have in this set. And what will be the size of array with five elements.

We have a pair a1-a1, a1- a2, a1-a3, a1-a4, a1-a5, a2-a2, a2-a3 etc. The pair a1-a2 and a2-a1 are equal due to symmetric. We also find some self-pairs (i.e. a1-a1, a2-a2 etc.). There are 25 pairs of elements, each of which is related or not (30 pairs – 5 self-pairs = 25). These are the total pairs and we may not have as much relations.

Q 116: Define the Union operation of disjoint set.

When we apply union operation on two sets, the members of both sets combined together and form a new set. In this case, there will be no duplicate entry in the new sets as these were disjoint. Union operation finds the set (root) of the element.

Q 117: Convert the given infix form to postfix form. Y-Z*X-Q^P

First will post power operator--->Y-Z*X-QP^

Then * operator will post-----> Y-ZX* - QP^

And then – operator will post-----> YZX*-QP^-

Q 118: Write Properties of equivalence class.

As a result of the reflexive, symmetric, and transitive properties, any equivalence relation provides a partition of the fundamental set into disjoint equivalence classes. Two elements of the given set are equivalent to each other if and only if they belong to the same equivalence class.

Q 119: Is this <= is an equivalence relation?

The relation “is less than or equal to”, denoted “ $\leq$ ”, is NOT an equivalence relation.

Q 120: How we insert new element in array at first position?

To insert an element in an array in C++ programming, you ask the user to enter the array size and array elements and ask for element with their position to insert.

Q 121: Write C++ code for recursive function call.

```
Voidrecurse () {
```

```
-----
```

```
Recurse ();
```

```
-----
```

```
}
```

```
Int main ()
```

```
-----
```

```
Recurse ();
```

```
}
```

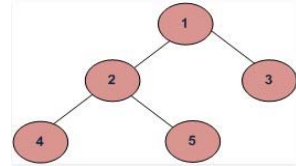
P-21 -----

Q 122: Write In order and preorder traversal.

Inorder Traversal

In case of binary search trees (BST), Inorder traversal gives nodes in non-decreasing order.

In Inorder we will go from (Left, Root, and Right). For example,



Example: Inorder traversal for the above-given figure is 4 2 5 1 3.

Preorder traversal

Preorder traversal is used to create a copy of the tree.

In Preorder we will go from (Root, Left, and Right). Same example, Preorder

(Root, Left, and Right): 1 2 4 5 3

Q 123: Show steps of merge sort:

11 12 13 21 22 23 31 33 41 42

Let we have two sorted arrays,

First we will compare elements of both arrays with each other one by one and then a small element will write in new additional array. 11 compare with 23.

11 elements are small, so we will put it in additional array and skip from array. 12 compare with 23. Small value put in additional array and skips.

13 with 23,

21 with 23,

22 with 23,

If we have no element remaining in first array then we will write all elements of second array as is it.

11	12	13	21	22	23	31	33	41	42
----	----	----	----	----	----	----	----	----	----

Q 124: For telephone directory which is best linear or non-linear array.

Linear array is best for telephone directory in data structure. Because a linear array is a list of finite number of same data type and all elements are stored in sequential memory location.

Q 125: Making Threaded binary tree using 40, 34, and 57,70,93,23 traversal orders should be Left to right.

See Concepts Only:

Q 126: Making skip list sketch s0 to s3Data are 12,34,56,78. Handout Page 442 See Concepts Only:

